Name
Date
Course

Trigonometric Equations

* When solving the following problems, do not use a calculator if your answer(s) are 0°, multiples of 30°, or multiples of 45°.

① Solve if $0 \le \theta < 360°$.

$\sqrt{2}\sin\theta + 1 = 0$

$\sqrt{2}\sin\theta = -1$

$\sin\theta = -\frac{1}{\sqrt{2}}$

$\boxed{\theta = 225°, 315°}$

② Find all values of w.

$-3 = \frac{\cot w}{4} - 3$

$0 = \frac{\cot w}{4}$

$0 = \cot w$

$\boxed{w = 90 + 180n, \text{ n is any integer.}}$

③ Solve if $0 \le A < 360°$.

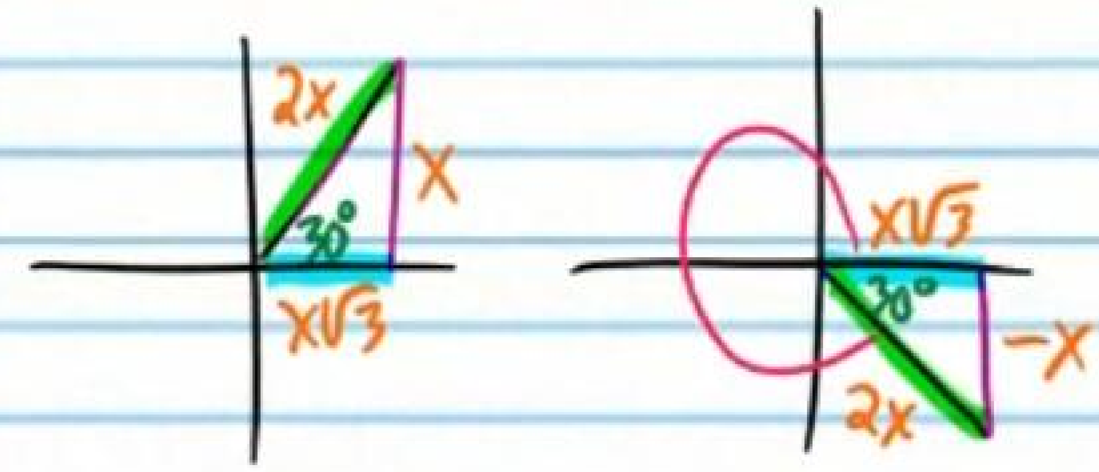
$7\sqrt{3}\sec A - 8 = 2 + 2\sqrt{3}\sec A$

$5\sqrt{3}\sec A - 8 = 2$

$5\sqrt{3}\sec A = 10$

$\frac{5\sqrt{3}\sec A}{5\sqrt{3}} = \frac{10}{5\sqrt{3}}$

$\sec A = \frac{2}{\sqrt{3}}$



$\boxed{A = 30°, 330°}$

④ Solve if $0 \le \beta < 360°$.

* Use a calculator.

$-2 + 3\tan\beta = 3 - 4\tan\beta$

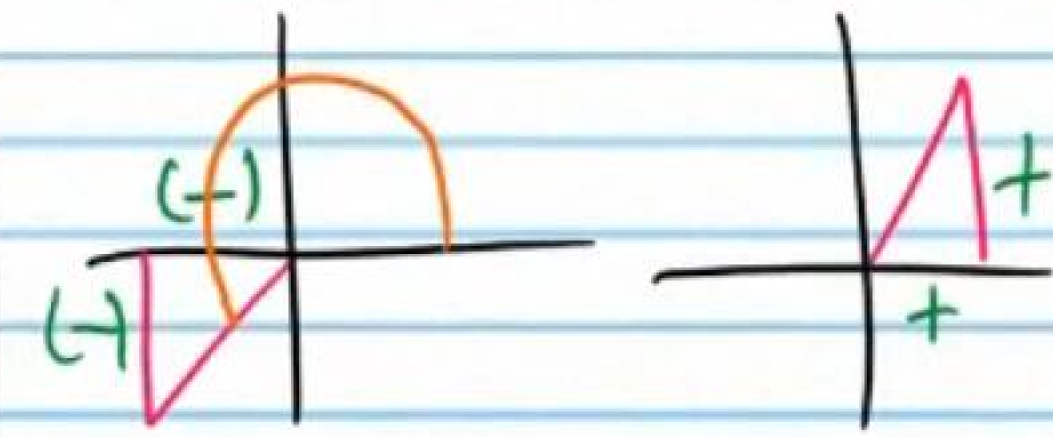
$-2 + 7\tan\beta = 3$

$7\tan\beta = 5$

$\tan\beta = \frac{5}{7}$

$\beta = \tan^{-1}\left(\frac{5}{7}\right)$

$\beta \approx 35.5°$



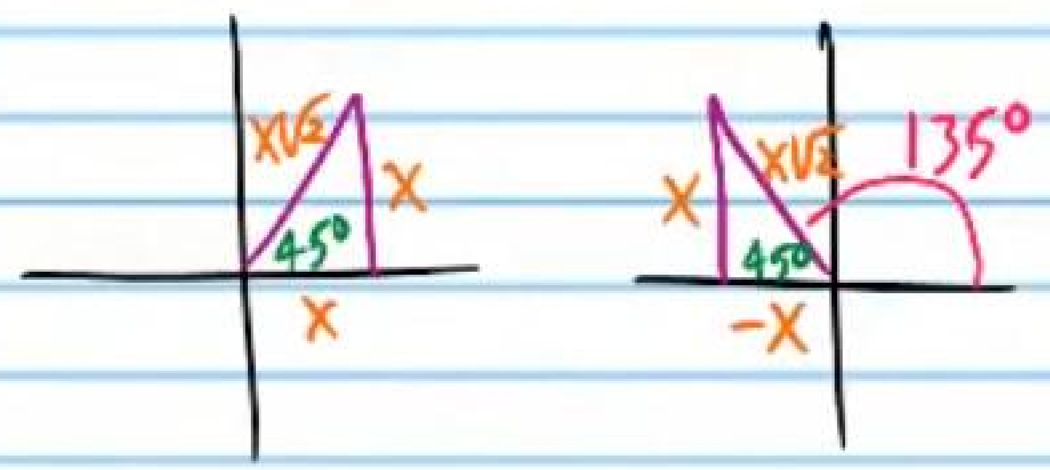
$\boxed{\beta \approx 35.5°, 215.5°}$

⑤ Solve if $0 \le \alpha < 360°$.

$4 + 5\csc\alpha = 5\sqrt{2} + 4$

$5\csc\alpha = 5\sqrt{2}$

$\csc\alpha = \sqrt{2}$



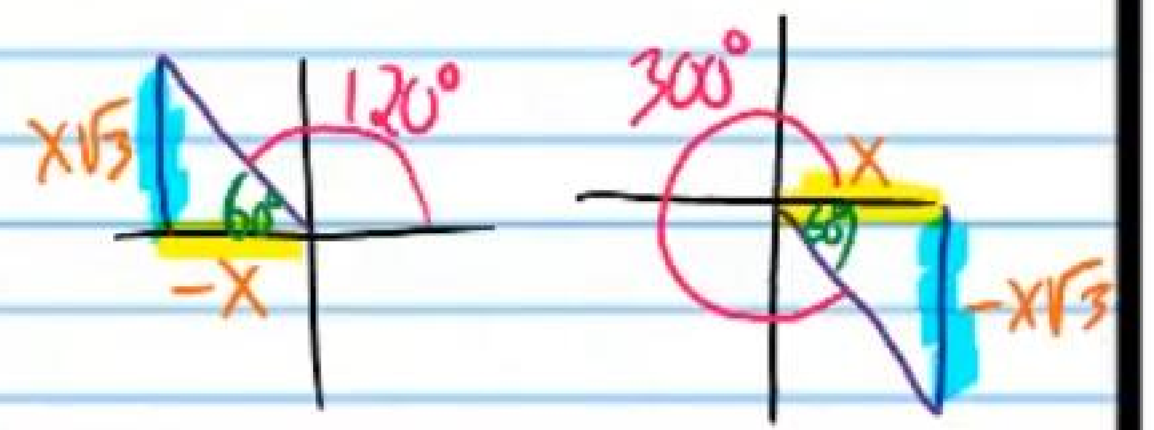
$\boxed{\alpha = 45°, 135°}$

⑥ Find all values of θ.

$-9\sqrt{3} + 3\tan\theta = -12\sqrt{3}$

$3\tan\theta = -3\sqrt{3}$

$\tan\theta = -\sqrt{3}$



$\theta = 120°, 300°$

$\boxed{\theta = 120 + 180n, \text{ n is any integer.}}$

⑦ Solve if $0 \le B < 360°$.

$-6\sec B - 7 = 1 - 14\sec B$

$8\sec B - 7 = 1$

$8\sec B = 8$

$\sec B = 1$

$\boxed{B = 0°}$

⑧ Solve if $0 \le W < 360°$

✱ Use a calculator, but remember that in this situation, most likely your calculator will not give you the reference angle.

$4\cos W - 1 = 8\cos W + 2$

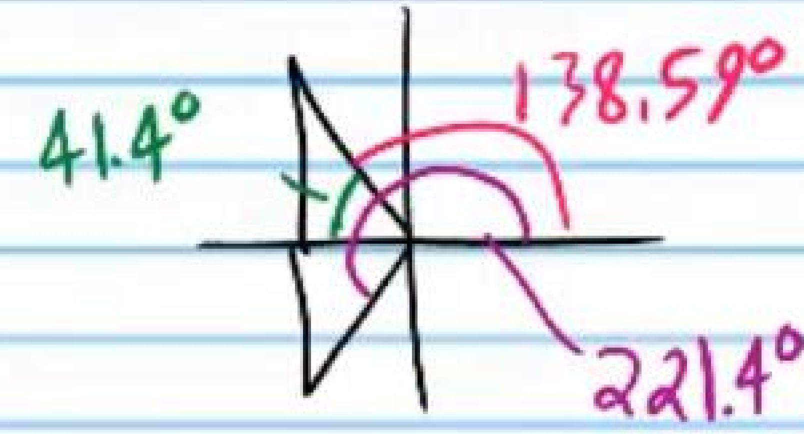
$-1 = 4\cos W + 2$

$-3 = 4\cos W$

$-\frac{3}{4} = \cos W$

$W = \cos^{-1}\left(-\frac{3}{4}\right)$

$\boxed{W \approx 138.59°, 221.4°}$



Using the Basic Identities and Factorization to Solve Equations

Using the Reciprocal Identities

⑨ Find all values of θ.

$\sin\theta = \csc\theta$

$\sin\theta = \dfrac{1}{\sin\theta}$

$\sin\theta \cdot \sin\theta = \dfrac{1}{\sin\theta} \cdot \sin\theta$

$\sin^2\theta = 1$

$\sqrt{\sin^2\theta} = \pm\sqrt{1}$

$\sin\theta = \pm 1$

$\theta = 90°, 270°, -90°, 450°$

$\boxed{\theta = 90 + 180n, \text{ n is any integer.}}$

⑩ Solve if $0 \le A \le 360°$

$3\cot A = \tan A$

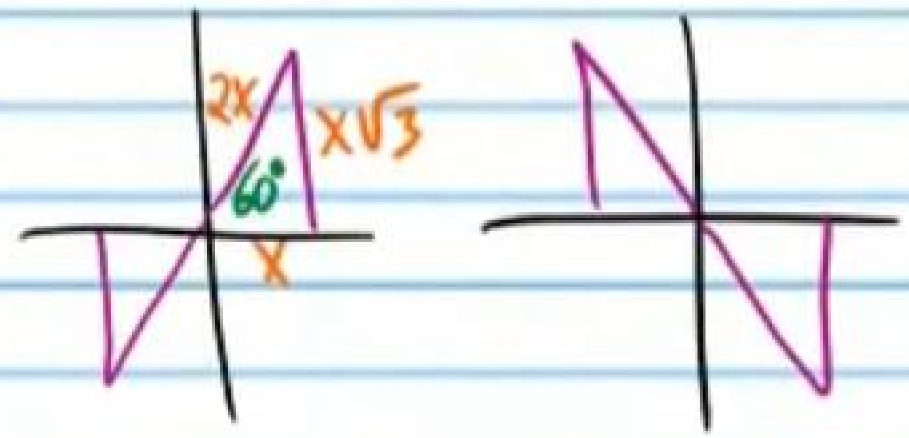
$3\dfrac{1}{\tan A} = \tan A$

$\cancel{\tan A} \cdot \dfrac{3}{\cancel{\tan A}} = \tan A \cdot \tan A$

$3 = \tan^2 A$

$\pm\sqrt{3} = \sqrt{\tan^2 A}$

$\pm\sqrt{3} = \tan A$



$\boxed{A = 60°, 120°, 240°, 300°}$

⑪ Solve if $0 \le B \le 360°$

★ Use a calculator.

$\sec B - 5\cos B = 0$

$\dfrac{1}{\cos B} - 5\cos B = 0$

$\cos B\left(\dfrac{1}{\cos B} - 5\cos B\right) = (0)\cos B$

$1 - 5\cos^2 B = 0$

$-5\cos^2 B = -1$

$\cos^2 B = \dfrac{1}{5}$

$\sqrt{\cos^2 B} = \pm\sqrt{\dfrac{1}{5}}$

$\cos B = \pm\dfrac{\sqrt{5}}{5}$

$\cos B = \dfrac{\sqrt{5}}{5}$ or $\cos B = -\dfrac{\sqrt{5}}{5}$

$B = \cos^{-1}\left(\dfrac{\sqrt{5}}{5}\right)$ or $B = \cos^{-1}\left(-\dfrac{\sqrt{5}}{5}\right)$

$\boxed{B \approx 63.43°, 296.57° \text{ or } B \approx 116.57°, 243.43°}$

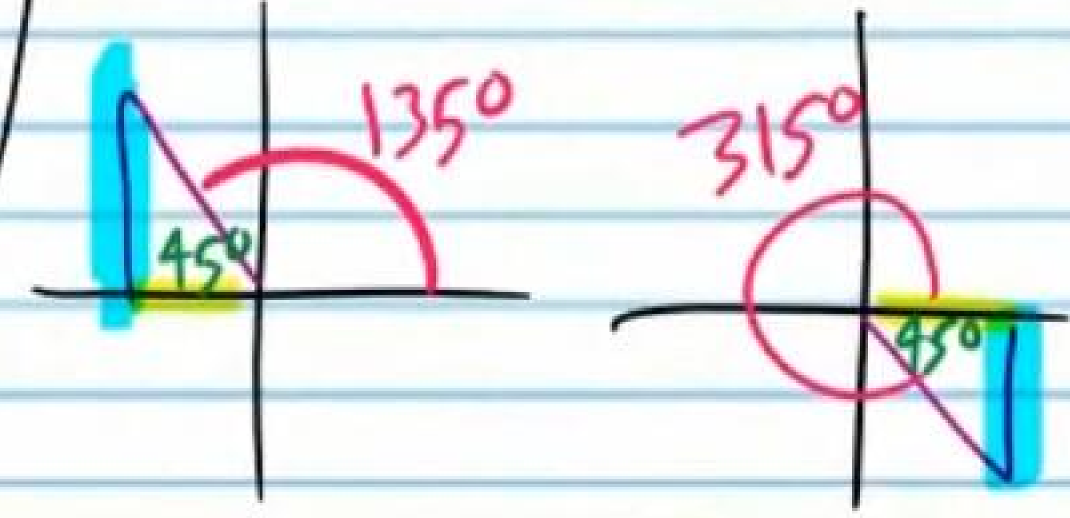
⑫ Solve if $0 < X < 360°$

$-4 = 5\tan X - \dfrac{1}{\cot X}$

$-4 = 5\tan x - \tan x$

$-4 = 4\tan x$

$-1 = \tan x$



$\boxed{X = 135°, 315°}$

⑬ Find all values of W.

✯ You need two equations.

$0 = 2\sin W - \csc W$

$0 = 2\sin W - \dfrac{1}{\sin W}$

$\sin W(0) = \left(2\sin W - \dfrac{1}{\sin W}\right)\sin W$

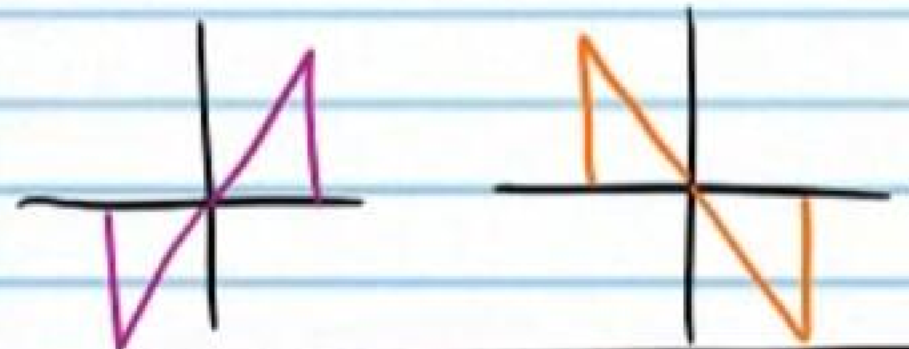
$0 = 2\sin^2 W - 1$

$1 = 2\sin^2 W$

$\dfrac{1}{2} = \sin^2 W$

$\pm\sqrt{\dfrac{1}{2}} = \sqrt{\sin^2 W}$

$\sin W = \pm\dfrac{1}{\sqrt{2}}$



$\boxed{\begin{array}{l} W = 45 + 180n,\ n \text{ is any integer} \\ W = 135 + 180n,\ n \text{ is any integer} \end{array}}$

⑭ Solve if $0 \le \theta \le 360°$

$\cos\theta = \sec\theta$

$\cos\theta = \dfrac{1}{\cos\theta}$

$\cos\theta \cdot \cos\theta = \dfrac{1}{\cos\theta} \cdot \cos\theta$

$\cos^2\theta = 1$

$\sqrt{\cos^2\theta} = \pm\sqrt{1}$

$\cos\theta = \pm 1$

$\boxed{\theta = 0°, 180°, 360°}$

⑮ Solve if $0 \le \alpha \le 360°$

✯ Use a calculator.

$\cot\alpha - 4\tan\alpha = 0$

$\dfrac{1}{\tan\alpha} - 4\tan\alpha = 0$

$\tan\alpha\left(\dfrac{1}{\tan\alpha} - 4\tan\alpha\right) = (0)\tan\alpha$

$1 - 4\tan^2\alpha = 0$

$-4\tan^2\alpha = -1$

$\tan^2\alpha = \dfrac{1}{4}$

$\sqrt{\tan^2\alpha} = \pm\sqrt{\dfrac{1}{4}}$

$\tan\alpha = \pm\dfrac{1}{2}$

$\tan\alpha = \dfrac{1}{2}$ or $\tan\alpha = -\dfrac{1}{2}$

$\alpha = \tan^{-1}\left(\dfrac{1}{2}\right)$ or $\alpha = \tan^{-1}\left(-\dfrac{1}{2}\right)$

$\alpha \approx 26.565°$ or $\alpha \approx -26.565°$

$\boxed{\alpha \approx 26.57°, 206.57° \text{ or } 153.43°, 333.43°}$

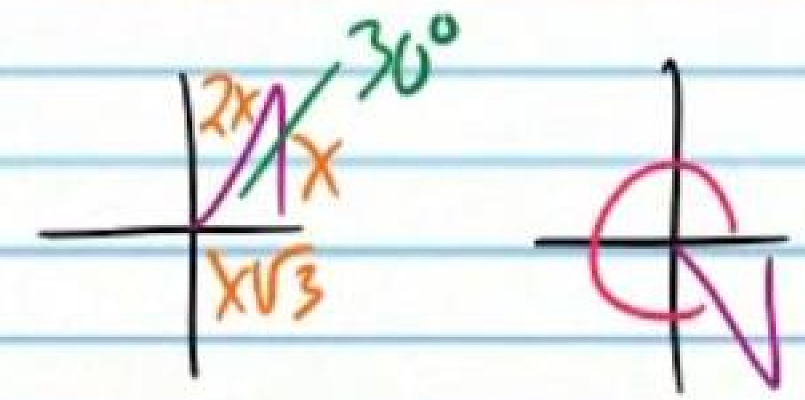
⑯ Solve if $0 < \beta < 360°$

$\dfrac{1}{\sec\beta} + \cos\beta = \sqrt{3}$

$\cos\beta + \cos\beta = \sqrt{3}$

$2\cos\beta = \sqrt{3}$

$\cos\beta = \dfrac{\sqrt{3}}{2}$



$\boxed{\beta = 30°, 330°}$

Using the Ratio Identities

(17) Find all values of y.

$\sin y = 2\tan y$

$\sin y = 2\dfrac{\sin y}{\cos y}$

$\cos y \cdot \sin y = 2\dfrac{\sin y}{\cos y}\cdot \cos y$

$\cos y \sin y = 2\sin y$

$\cos y \sin y - 2\sin y = 2\sin y - 2\sin y$

$\cos y \sin y - 2\sin y = 0$

$\sin y(\cos y - 2) = 0$

$\sin y = 0$ or $\cos y - 2 = 0$

$\boxed{y = 180n,\ n \text{ is any integer}}$ or $\cos y \neq 2$

(18) Solve if $0 \le A \le 360°$

$\sqrt{3}\sin A + \cos A = 0$

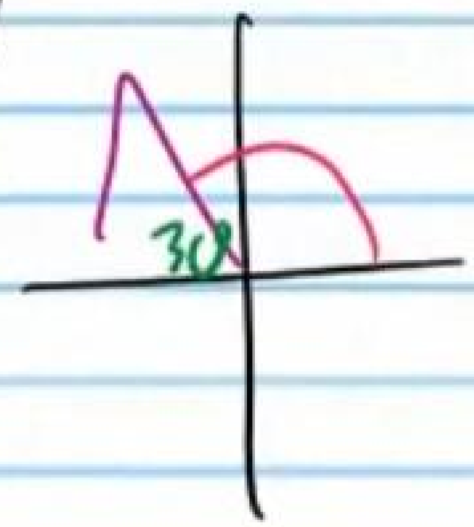
$\sqrt{3}\sin A + \cos A - \cos A = 0 - \cos A$

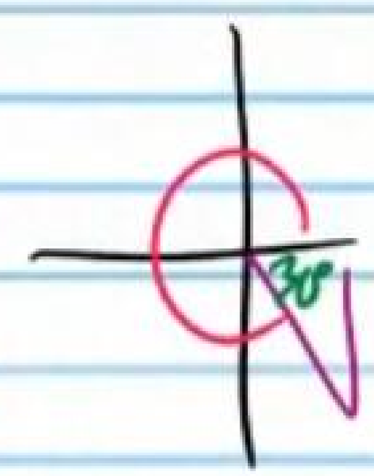
$\sqrt{3}\sin A = -\cos A$

$\dfrac{\sqrt{3}\sin A}{\cos A} = \dfrac{-\cos A}{\cos A}$

$\sqrt{3}\tan A = -1$

$\tan A = -\dfrac{1}{\sqrt{3}}$





$\boxed{A = 150°, 330°}$

(19) Solve if $0 \le \theta < 360°$

$\cos\theta = -\cot\theta$

$\cos\theta = -\dfrac{\cos\theta}{\sin\theta}$

$\sin\theta\cos\theta = -\dfrac{\cos\theta}{\sin\theta}\cdot\sin\theta$

$\sin\theta\cos\theta = -\cos\theta$

$\cos\theta + \sin\theta\cos\theta = -\cos\theta + \cos\theta$

$\cos\theta + \sin\theta\cos\theta = 0$

$\cos\theta(1 + \sin\theta) = 0$

$\cos\theta = 0$ or $1 + \sin\theta = 0$

$\theta = 90°, 270°$ or $\sin\theta = -1$

$\theta = 270°$

$\boxed{\theta = 90°, 270°}$

(20) Find all values of B.

$\cos B = \sin B$

$\dfrac{\cos B}{\cos B} = \dfrac{\sin B}{\cos B}$

$1 = \tan B$

$B = 45°, 225°$

$\boxed{B = 45 + 180n,\ n \text{ is any integer.}}$

Using the Pythagorean Identities

(21) Solve if $0 < \beta < 360°$.

$1 + \cot\beta = \csc^2\beta$

$1 + \cot\beta = 1 + \cot^2\beta$

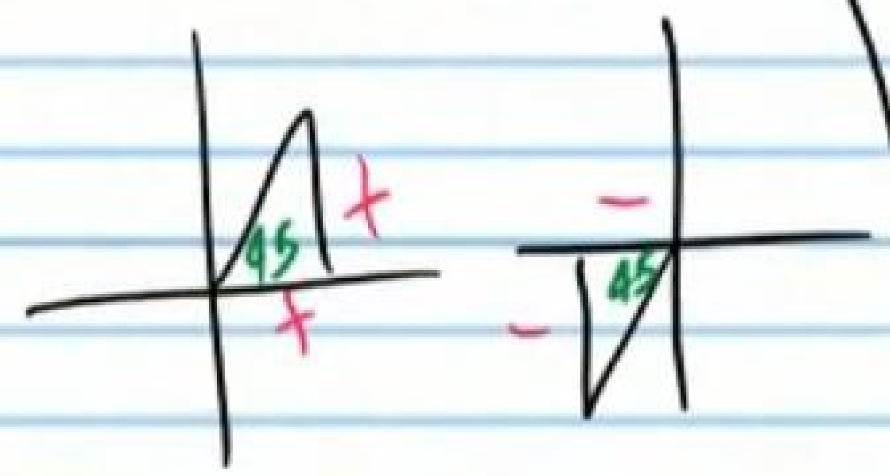
$\cot\beta = \cot^2\beta$

$0 = \cot^2\beta - \cot\beta$

$0 = \cot\beta(\cot\beta - 1)$

$\cot\beta = 0$ or $\cot\beta - 1 = 0$

$\beta = 90°, 270°$ $\qquad$ $\cot\beta = 1$



$\beta = 45°, 225°$

$\boxed{\beta = 45°, 90°, 225°, 270°}$

(22) Solve if $0 < \alpha < 360°$.

$\sec^2\alpha + \cos\alpha = \tan^2\alpha$

$\tan^2\alpha + 1 + \cos\alpha = \tan^2\alpha$

$1 + \cos\alpha = 0$

$\cos\alpha = -1$

$\boxed{\alpha = 180°}$

(23) Solve if $0 \le \theta \le 360°$

$\sin\theta + \cos\theta = 1$

$(\sin\theta + \cos\theta)^2 = 1^2$

$(\sin\theta + \cos\theta)(\sin\theta + \cos\theta) = 1$

$\sin^2\theta + \sin\theta\cos\theta + \cos\theta\sin\theta + \cos^2\theta = 1$

$\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 1$

$1 + 2\sin\theta\cos\theta = 1$

$2\sin\theta\cos\theta = 0$

$\sin\theta\cos\theta = 0$

$\sin\theta = 0$ or $\cos\theta = 0$

$\theta = 0°, 180°, 360°$ or $\theta = 90°, 270°$

$\theta \neq 180°, 270°$

$\boxed{\theta = 0°, 90°, 360°}$

(24) Solve if $0 < w < 360°$

$1 = \cos w - \sin w$

$1^2 = (\cos w - \sin w)^2$

$1 = (\cos w - \sin w)(\cos w - \sin w)$

$1 = \cos^2 w - \sin w\cos w - \sin w\cos w + \sin^2 w$

$1 = \sin^2 w + \cos^2 w - 2\sin w\cos w$

$1 = 1 - 2\sin w\cos w$

$0 = -2\sin w\cos w$

$0 = \sin w\cos w$

$\sin\theta = 0$ or $\cos\theta = 0$

$\theta = 0°, 180°, 360°$ or $\theta = 90°, 270°$

$\theta \neq 90°, 180°$

$\boxed{\theta = 270°}$